

Shreyas Narayanan

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EDUCATION

University of Michigan College of Engineering

B.S.E. in Computer Engineering | GPA: 3.67

Ann Arbor, MI

Expected December 2027

- **Intending to pursue M.S.E. in ECE with focus in Computer Architecture.**
- **Relevant Coursework:** Parallel Computer Architecture, Microarchitecture, Computer Architecture, GPU Programming in CUDA, Digital IC Design, Embedded Systems, Data Structures and Algorithms

TECHNICAL SKILLS

Programming: SystemVerilog, C/C++, Python, Assembly (ARM/RISC-V), Tcl, PyTorch, CUDA

Tools: Cadence Virtuoso, Synopsys (Verdi, VCS), Quartus Prime, ModelSim, Altium

Platforms: AXI, MESI/MOESI, SystemC, TLM, MCUs (STM, Microchip), FPGAs (Altera), SoC, UVM, FreeRTOS, JTAG

EXPERIENCE

RISC-M Computer Architecture Club

April 2026 – Present

Project Engineer

- Co-founded UMich's first Computer Architecture organization, directing team-wide RTL and Physical Design projects while connecting peers with industry leaders. Grew the club to over 40 members and 3 ongoing projects.
- Leading a team to implement the cycle-approximate simulator using systemC and TLM while modeling the in-order scalar core and vector lane arbitration logic to benchmark collective operations against GPU-accelerated training and inference.

Reditus Space

May 2026 – August 2026

Flight Software Intern

- Implemented a CANopen driver on STM32 FDCAN with segmented SDO transfers and priority-queued command/telemetry scheduling, connecting it through PropManager to a full propulsion pipeline validated through HITL testing.
- Extended GenericPayload with upload, download, script execution, and file-downlink commands over RCM payload firmware, validating resource lifecycle, GDS serialization, and multi-chunk transfer limits across a full integration test suite
- Built indexed firmware image validation and transactional SD-card publication across the bootloader/flight-software boundary, adding CRC sidecar generation, incremental scrubbing, and FRAM-backed boot-record recovery.

Interactive Sensing and Computing Lab

January 2025 – Present

Embedded Systems Engineering Intern

- Ported a low-power voice obfuscation algorithm from Python to C on an STM32G431KB, implementing Q31 fixed-point arithmetic and 6-band frequency recombination under tight RAM and I2S-to-DAC bit-width constraints.
- Validated obfuscation fidelity by reverse-engineering an Amazon Echo, soldering an analog injection line to the mic PCB, and driving it via DAC across 900 test conditions, preserving 88% of baseline Alexa recognition accuracy.
- Designed a real-time magnetic signature detection algorithm in C, leveraging digital signal processing to classify eyedropper cap movements across multiple states via a threshold-based state machine and achieving a 95% detection accuracy.
- Developed a fiber laser PCB prototyping process, achieving 0.25mm accuracy and cutting iteration time to 1.5 hours.

University of Michigan College of Engineering

August 2025 – December 2025

Instructional Aide for Introduction to Logic Design

- Taught weekly labs for 30+ undergraduates, instructing on Verilog RTL modeling, FPGA synthesis, and digital logic.
- Mentored 20+ students via office hours and Piazza, guiding HDL debugging, RTL best practices, and core concepts.

PROJECTS

P6-Style RTOS-Capable RISC-V Processor

- Co-designed a 2-way superscalar P6-style OoO RISC-V processor in SystemVerilog featuring distributed reservation stations, dual CDBs, and an ROB to guarantee precise in-order retirement and rapid branch recovery.
- Engineered instruction and data stream prefetchers alongside MMIO decoding logic, interfacing with a writeback L1 cache, victim cache, and conservative LSQ to minimize memory stalls.
- Implemented RV32IM + Zicsr ISA support, developing the trap handling state machine, atomic CSRs, and CLINT hardware timer to enable preemptive scheduling and rapid context switching for FreeRTOS.

Predictive CPU-Accelerator Coherence over CXL

- Building a region level sharing predictor in gem5 and SimCXL that classifies memory regions as private or actively shared from request history and ownership changes, cutting CXL coherence traffic on CPU-accelerator workloads.
- Designing the coherence mode state machine that moves predicted private regions to a low overhead mode and forces a safe transition back to shared before any conflicting access completes so mispredictions cost performance but not correctness
- Evaluating against an always coherent baseline and an oracle predictor across CXL latency/bandwidth, region size, sharing frequency, and private-phase duration, measuring link traffic, stall time and misprediction penalty